xMCF
ISO/PAS 8329

Description of mechanical connections and joints in structural systems

ISO/PAS 8329 xMCF WG meeting

focus group on issues #105 / #61

2026-01-28

Meeting results are added in green font.

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energizing great minds

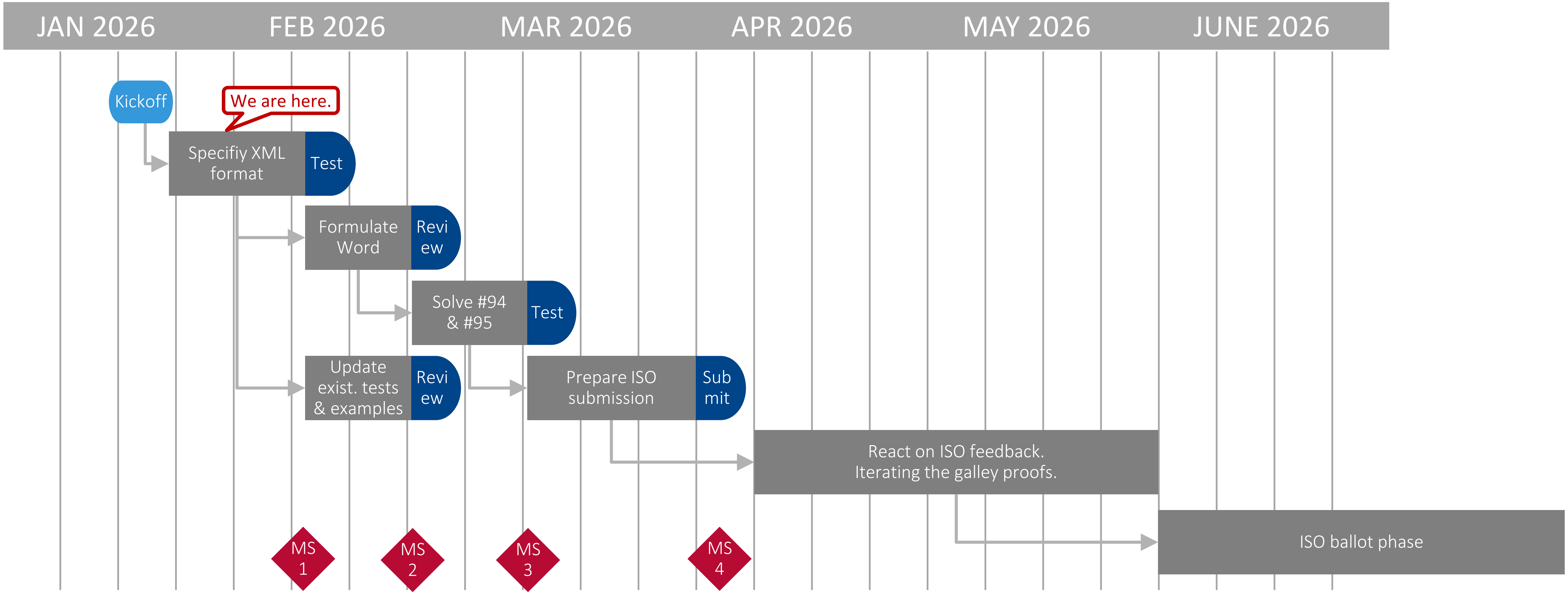
Agenda

As announced at https://github.com/economidis-nick/createXSDforxMCF/blob/master/VDA_FAT_AK_25/Meetings/2026-01-28_VideoConference/README.md

Today's objective: **Create XML format suggestions**

- Welcome
- Timeline for next release
- Focus on detailing the XML elements, based e.g. on the comment in GitHub at issue #105.
- Steps we should take next

Timeline for a release in the summer



Updating the PAS (or else striving for an IS, directly) shall be cross-checked with the chair of ISO/TC 184 / SC 4.
Last week’s meeting of the Policy and Planning Committee (PPC) was postponed to 18th of Feb.

Detailing XML Elements — Template Syntax, Screw Example

We identified the example XML file chapter_9_5_7_exampleA__patched_for_xMCF3.2.xml (provided by the comment of 2026-10-14 09:17+0100) as a good starting point.

It suggests as a syntax for defining an (file-embedded) list of *connection templates* at XML root level:

```
<connection_templates>
    <connection_0d label="screw_template_4711">
        <threaded_connection length="50." diameter="10"
            head_diameter="16.0" head_height="5" sink_size="4" thread_length="35"
            torque="80" angle="30" pretension="1200" part_code="M10x50 12.9" >
            <screw /> <!-- Screw may come without any attributes -->
            <washer outer_diameter="20"/>
        </threaded_connection>
    </connection_0d>
</connection_templates>
```

Homework: Consider what “misuse” or contradictory set of data can occur by this approach!

Detailing XML Elements — Reference Syntax, Screw Example

The example suggests as syntax for *referring to* a connection template at <connection_0d/> level:

Default form — no change to / overwriting template data:

```
<!-- Screw exactly as defined in connection_templates: -->
<connection_0d label="SCREW_100532" template="screw_template_4711">
    <threaded_connection>
        <normal_direction x="2.9" y="0.1" z="0.0" />
    </threaded_connection>
    <loc> 1500.3809 838.75885 730.6529 </loc>
</connection_0d>
```

Minimal change — overwriting an attribute value:

```
<!-- Screw exactly as defined in connection_templates, but with smaller torque: -->
<connection_0d label="SCREW_100533" template="screw_template_4711">
    <threaded_connection torque="70" >
        <normal_direction x="3.0" y="0.0" z="0.0" />
    </threaded_connection>
    <loc> 1540.3809 838.75885 730.6529 </loc>
</connection_0d>
```

Detailing XML Elements — Reference Syntax, Screw Example

The example suggests as syntax for *referring to* a connection template at `<connection_0d/>` level:

<u>Overwriting a child element:</u>

```
<!-- Screw exactly as defined in connection_templates, but with larger washer: -->
<connection_0d label="SCREW_101537" template="screw_template_4711">
    <threaded_connection>
        <normal_direction x="3.1" y="-0.1" z="0.0" />
        <washer outer_diameter="30"/>
    </threaded_connection>
    <loc> 1580.3809 838.75885 730.6529 </loc>
</connection_0d>
```

Attention: Here, only the `outer_diameter` of the `<washer/>` is overwritten. Any other washer-parameter given from the template stays effective.

Detailing XML Elements — Template Syntax, Bolt Example

The example extends to cases of *deeper structured* elements:

```
<connection_templates>
    <connection_0d label="bolt_template_4712">
        <threaded_connection length="50" diameter="10"
                             head_diameter="16" head_height="5" thread_length="35"
                             torque="80" angle="30" pretension="1200" part_code="M10x50 12.9">
            <normal_direction x="0" y="0" z="-10"/>
            <washer outer_diameter="20" inner_diameter="10.3" thickness="1.5" attached="true"/>
            <bolt>
                <nut diameter="16." height="5" static_friction="0.8" clipped_to="4"/>
            </bolt>
        </threaded_connection>
    </connection_0d>
</connection_templates>
```


Detailing XML Elements — Reference Syntax, Bolt Example

The connection template can be *referred to* for example like this:

```
<!-- Bolt exactly as defined in connection_templates, but with less torque, larger washer & nut: -->
<connection_0d label="BOLT_100732" template="bolt_template_4712">
    <threaded_connection torque="70">
        <normal_direction x="3.1" y="-0.1" z="0.0"/>
        <washer outer_diameter="30"/>
```

Check, whether a washer may refer to a connection template, too! → we need examples to better understand.

```
        <bolt>
            <nut height="8"/>
```

Check, whether a nut may refer to a connection template, too! → we need examples to better understand

```
        </bolt>
    </threaded_connection>
</connection_0d>
```

Rule (→ Word file): If the connection template is a `<screw/>`, the referring element must not redefine it to a `<bolt/>` and vice versa!
To be extended to similar situations in case of future extensions of χMCF, e.g. a spotweld (from template) must not be redefined to a gumdrop!

Detailing XML Elements — Template Syntax, Gumdrop Example

(new — see file “chapter_10_5_exampleB__with_templates.xml”)

For a gumdrop, the example template would be:

```xml
<connection_templates>
    <connection_0d label="GumDrop template">
        <!-- Assumed Unit system with mass attribute with value="kg" -->
        <gumdrop diameter="5.0" mass="0.0033" material="Glue_Material" />
    </connection_0d>
</connection_templates>
```

Detailing XML Elements — Reference Syntax, Gumdrop Sequence

(new — see file “chapter_10_5_exampleB__with_templates.xml”)

For a gumdrop sequence, the example reference could be (as suggested in GitHub comment):

```
<!-- Sequence of gum drops exactly as defined in connection_templates: -->
<connection_1d label="DROP_LINE_11000" template="GumDrop template">
    <sequence_connection_0d spacing="30.0" margin="1.0" priority="spacing">
        <gumdrop/>
    </sequence_connection_0d>
    <loc_list>
        …
    </loc_list>
</connection_1d>
```

Observation (CF): A template for a 0d item as parameter of a 1d item appears inappropriate to me.

What about this approach:

```
<connection_1d label="DROP_LINE_11000" template="Later, here could be place for a sequence template.">
    <sequence_connection_0d spacing="30.0" margin="1.0" priority="spacing" template="GumDrop template">
```


XML Example File provided by @ghirel / PDM-IF.

The example “SAMPLE-FASTENER_003--.--1In Work000-ACOMMON_ENG_01_xmcf3.2 GHi.xml” has shown us that:

- <loc/> must not appear inside <connection_templates/>.
- <stacking/> must not appear inside <connection_templates/>.

Contentual Action Items ≙ Next Steps:

- **Standard document (Word file):**
  - Description of overall concept & objectives, intended application & motivation etc.
  - Description of semantics for internal catalog
  - Description of semantics for reference to a connection template
  - Definition of example use cases and in-document XML-examples
- **XML & XSD** — See discussion in #105!
  - Sketching XML example file(s), incl. (w.i.p.)
    - precedence rule (w.i.p.) and dropped mandatory attribute,
    - internal (w.i.p.) and external file-based catalog
  - Description of syntax for internal catalogs (w.i.p.)
  - Description of syntax for reference to a connection template:
    - internal (w.i.p.), external file-based, external database-based
  - Drafting XML schema (XSD) and test cases / test example files
    Homework: Collect use cases which can yield e.g. test cases, shared as Markdown files in GitHub at https://github.com/economidis-nick/createXSDforxMCF/tree/issues_61_94_95_105/VDA_FAT_AK_25/Meetings/2026-01-28_VideoConference
  - Investigate χMCF elements and attributes for eligibility for template.
    ⇒ we need a rule, here! ⇒ Homework!

Next meetings

Conclusion from this discussion: …

- 2026-02-04, 15:00 CET — focus group on #105 / #61 (*repurposed time slot*)
  — objective: agree on XML format suggestions
- 2026-02-11, 15:00 CET — focus group on #105 / #61 (*repurposed time slot*) collides with PDM-IF Workshop
  — objective: agree on XML format suggestions
- 2026-02-18, 15:00 CET — full group (*repurposed time slot*) collides with ISO TC 184 SC 4 PPC
  — objective: agree on XML format suggestions
- 2026-MM-DD, HH:MM CET — full group (*new time slot to be found*)
  — objective: agree on DOC suggestions & examples

CF sent / will send out invitations.